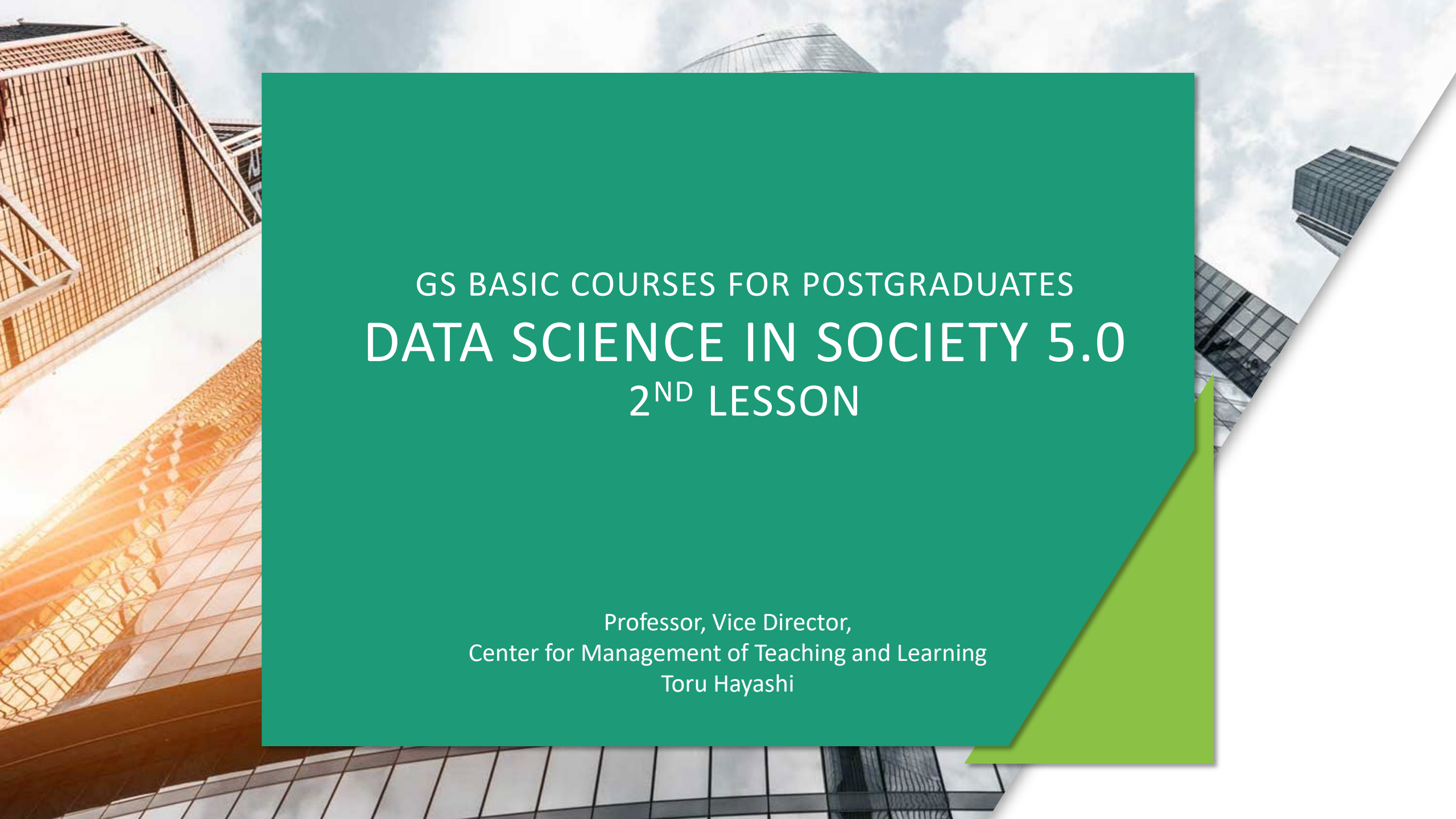


The background of the slide features a low-angle photograph of modern skyscrapers. On the left, a building with a distinctive orange-brown, grid-like facade is visible. To the right, a glass-clad skyscraper reaches towards a sky filled with soft, white clouds. A large, semi-transparent green rectangle is positioned in the center, serving as a backdrop for the text. The bottom right corner of this green rectangle is folded over, revealing a lighter green layer underneath.

GS BASIC COURSES FOR POSTGRADUATES

DATA SCIENCE IN SOCIETY 5.0

2ND LESSON

Professor, Vice Director,
Center for Management of Teaching and Learning
Toru Hayashi

2ND LESSON THEME

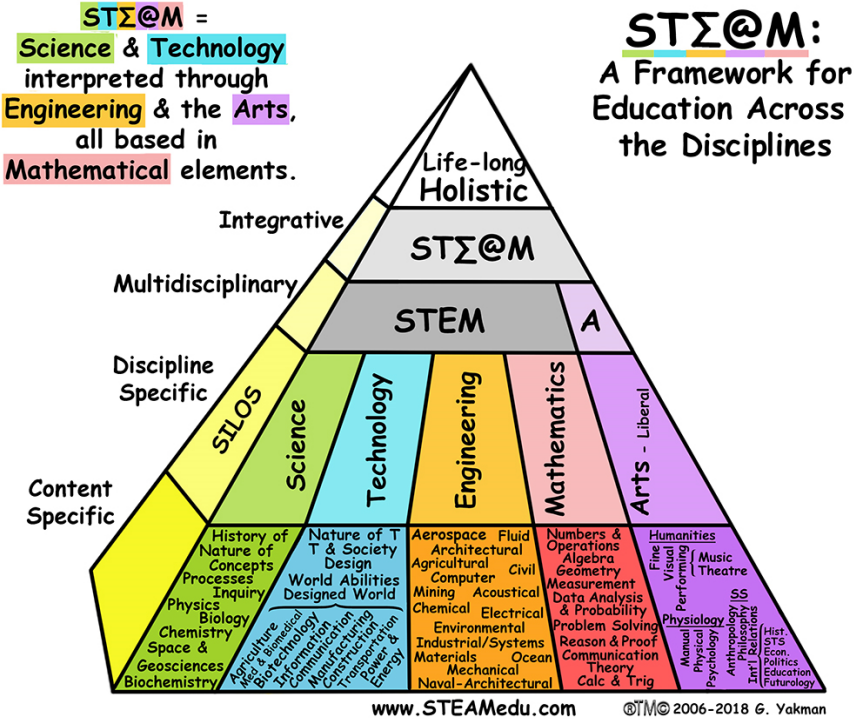
"SIGNIFICANCE AND VALUE OF STEAM EDUCATION AND
INTEGRATED KNOWLEDGE IN A KNOWLEDGE INTENSIVE
SOCIETY"

PART 1: SIGNIFICANCE AND VALUE OF STEAM EDUCATION

1] STEAM education as a fusion of arts and sciences and a cross-disciplinary approach

The program combines STEM education (Science, Technology, Engineering, and Mathematics) with the creative element of Art (Art and Design) to develop human resources with a bird's eye view and practical ability to apply what they learn in each field to solve problems in the real world. The aim is to nurture human resources who have acquired the ability to look over and put into practice the knowledge they have acquired in each area of study.

STEAM is "Science & Technology, interpreted through Engineering & the Arts, all based in mathematical elements." (Science & Technology, interpreted through Engineering & the Arts, all based in Mathematical elements)



area	concrete details
science	History of Science, Nature of Science, Physics, Biology, Chemistry, Space Science, Earth Science, Biochemistry
technology	Nature of technology, technology and society, design, global skills, designed world (agriculture, biomedicine, biotechnology, telecommunications, manufacturing, transportation, workforce and energy)
engineering	Aerospace, fluids, architecture, agriculture, civil engineering, computers, mining, acoustics, chemical engineering, electronics, environment, industrial systems, materials, marine, mechanical engineering, shipbuilding
mathematics	Number and operations, algebra, geometry, surveying, data analysis and probability, problem solving, reasoning and proof, communication theory, computation and trigonometry
arts	Humanities (arts, visual arts, performing arts (music, theater)), social sciences (anthropology, philosophy, international relations, history, science and technology, economics, political science, education, futurology), functions and mechanisms of the human body (manual, physical, psychological)

From STEM to STEAM

Please click on the URL
below to watch the video.

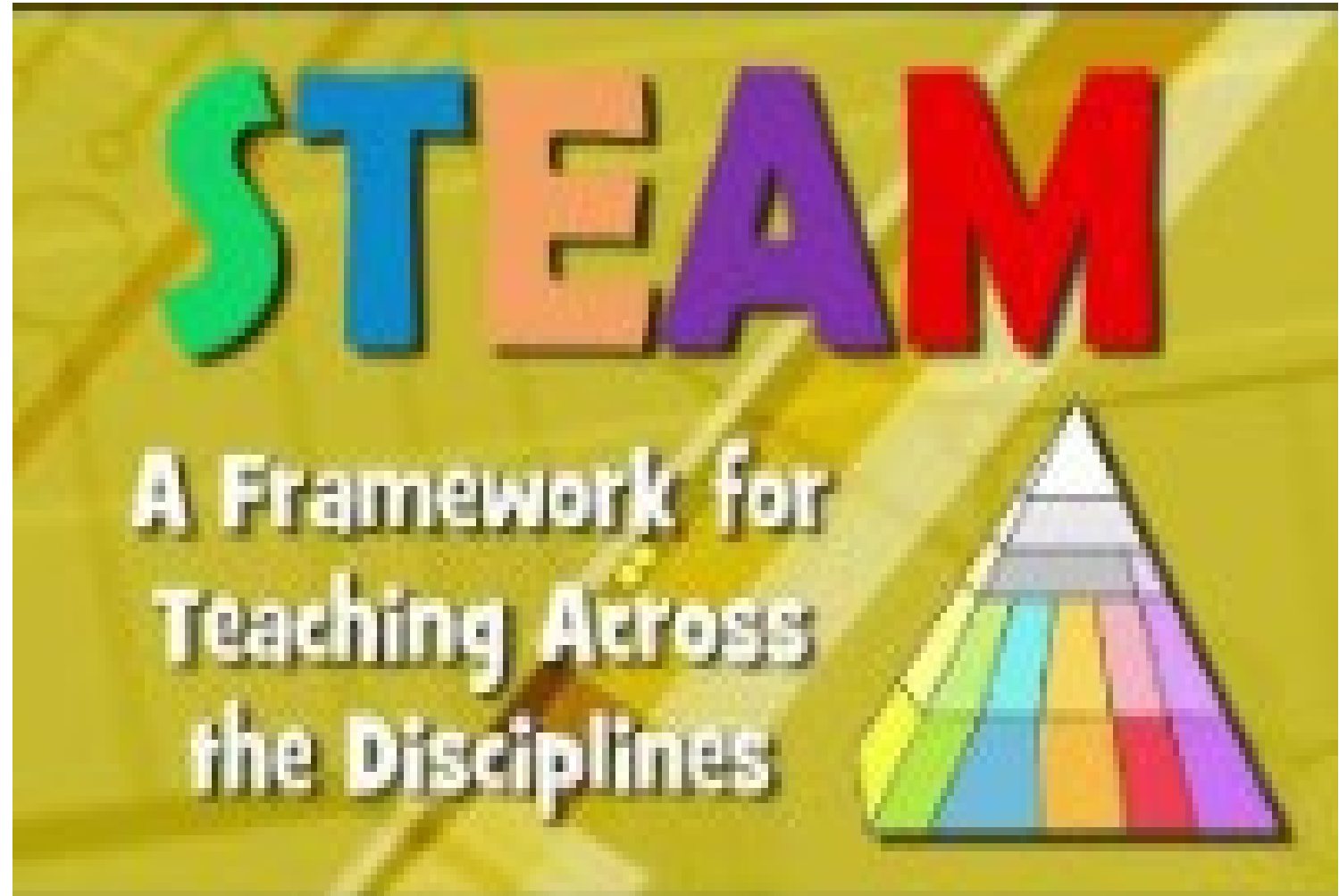
<https://youtu.be/6XvmhE1J9PY>



Framework and Value of STEAM Education

Please click on the URL
below to watch the video.

<https://youtu.be/0tkAvWWhKf4>



Background and History : The Common Theme of Active Learning

Summary of Report: "Towards a Qualitative Transformation of University Education for Building a New Future - Universities Fostering Lifelong Learning and the Ability to Think Independently and Proactively"

1. Role of University and Purpose of Report

Living in an unpredictable era

- ◆ Japan is affected in various ways by a rapidly changing society including: globalization, advanced information society, rapidly aging population and declining birthrate.



Universities faced with greater demands and expectations

- ◆ Businesses and communities have high expectations for human resources and academic research which help to pioneer an unpredictable era.
- ◆ Higher education has entered a new stage with the university entrance rate exceeding 50 percent.
- ◆ A decade has passed since the establishment of national university corporations, and implementation of the approval assessment system.

Society demands human resources with the intellectual and conceptual ability to paint a future vision for society. Universities are autonomous centers of knowledge creation and accumulation, and must cultivate individuals who take leadership and help create a future vision of society. Universities are called to:

- define a vision for future society and the role of universities based on new knowledge and ideas
- cultivate human resources capable of becoming leaders of the next generation, promote academic research

2. Basic points examined

Interactive exchanges among various parties, importance of objective data

How to cultivate knowledge through lower/upper secondary school and higher education

Speedy reform necessary

3. Vision for Society and required capabilities

Vision for Society

- ◆ A mature society that actively draws on the outstanding knowledge and ideas to develop, and that continues to maintain fair and stable growth.
⇒ A model of independence, collaboration and creativity underpinned by knowledge



Required capabilities in a mature society

- ◆ Cognitive capabilities, including critical and rational thinking, to identify solutions to "unsolvable" issues.
- ◆ Ethical and social capabilities to perform duties, demonstrate teamwork and leadership, and fulfill social responsibilities.
- ◆ Creative and conceptual capabilities based on extensive and ongoing academic experience.
- ◆ "Gakushiryoku" - ability required for university graduates for an unpredictable era including the education, knowledge and experience to make correct decisions in the face of unexpected difficulties.

4. Qualitative transformation needed for undergraduate education

- ◆ Undergraduate education must enhance subjective learning. The cultivation of "gakushiryoku" requires active learning through discussions and debates, and experiential education programs through internships.
- ◆ Students can acquire abilities for lifelong learning and to think independently only through gathering experiences in active learning. In order to do so, securing quality study time is needed.



5. Study time of university students

- ◆ Short study time (students study an average 4.6 hours/day during the school term).
- ◆ Citizens, businesses and students unsatisfied with current university education.
- ◆ Universities (presidents, deans, students) unsatisfied with students' general knowledge and study time outside of classroom.
- ◆ Study time for high school students with mid-level academic ability has decreased by nearly half over the last 15 years.

Therefore

6. Methods for universities to advance qualitative transformation of higher education

- ◆ Necessary measures to reach a qualitative turning point to increase and secure students study time
• Structuring curriculums (organize and consolidate class subjects) • Providing organized education • Improving curriculums (syllabus) • Consolidation and coordination between syllabuses and subjects
- ◆ Shift from teacher-centered composition of class subjects to establishing curriculum-based education for bachelors degrees

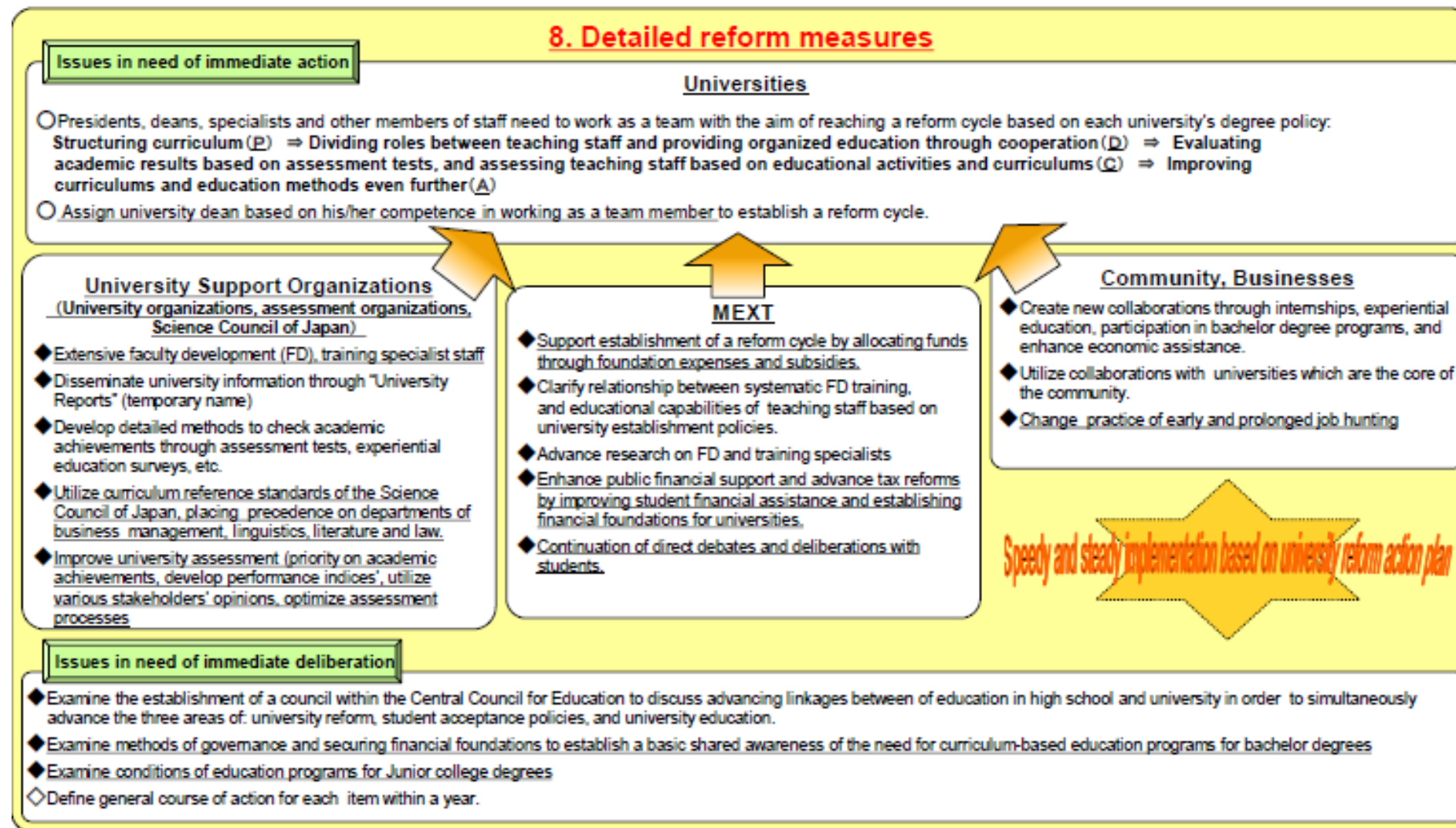
Issues

7. Issues towards advancing qualitative transformation of higher education

1. Lack of basic awareness on the concept of "curriculum-based education programs for bachelor's degrees"
2. Further improvements needed in learning-support environment
3. Improving linkage and collaboration between education in high schools and universities
4. Improving linkage between society and universities (change practice of students' early and prolonged job hunting)

To advance qualitative transformation of university education by clearing the above

Background and History :The Common Theme of Active Learning



Organizational Efforts in STEAM Education at Universities

Related Efforts in Recent Years

Faculty of Co-Creation, Kyushu University

Department of International Co-creation, Faculty of Integrated Sciences, Hiroshima University

Tokyo Institute of Technology, School of Liberal Arts Research and Education

Kagawa University DRI Education

University-wide STEAM education efforts

Kanazawa University, Kanazawa University

Leading STEAM Human Resource Development

Program

Common Education GS Courses

Graduate School GS Courses

【本事業で取り組む4つの実施項目】

事業
目標

融合した専門知と鋭敏な飛躍知を持つ

社会変革先導人材の育成



定員55名

① 新しい学域を設置
イノベーション &アントレプレナーシップ

融合学域 先導学類

人間社会学域 理工学域 医薬保健学域
に続く、新しい学域

定員30名

② 新しいプログラムを開講
異分野融合・異文化協働

**先導STEAM人材
育成プログラム**

全学域の学生 対象の文理融合プログラム

③ 新しいリベラルアーツ教育・STEAM教育の展開

全学士課程学生が対象

リベラルアーツ教育の大幅拡充

STEAM教育の全学必修化
共通教育GS科目第6群を新設

④ 教学マネジメントセンターを設置

教育改革の司令塔

① 新しい学域
② 新しいプログラム

③ リベラルアーツ教育
STEAM教育 を含め

教学マネジメントを運営・支援

- ① 融合学域の設置 => R3学域と先導学類設置、R4観光デザイン学類設置、R5スマート創成科学類設置申請
・融合学域の志願倍率（前期日程・一般選抜ー本学は後期日程なし）
先導学類：初年度3.5倍、2年目3.0倍、観光デザイン学類：初年度3.0倍（いずれも学内屈指）
- ② 先導STEAM人材育成プログラム（KU-STEAM）の新設
・計画前倒しで、R3後期から導入科目3科目開講・約100名受講、ラーニング・コミュニティ形成の整備
・R4全学域対象プログラム導入科目「融合先導知実践演習」の抽選倍率2倍超
=>事業参画学生：R3実績1,864名、R4実績3,031名（1年次1,065名、2年次以上1,966名）
・R4参画教員を68名に拡充 ・パンフを配布、履修ガイダンスや説明会を連続的に開催
- ③ 学士課程のリベラルアーツ科目である共通教育GS科目と学域GS専門科目の拡充
・数理・データサイエンス・AI教育、デザイン思考等、新しいリベラルアーツ教育を全学展開
・共通教育GS科目第6群は延べ3,917名が受講 ・大学院全研究科へ必修科目配置し接続

Issues often pointed out in education that integrates the humanities and sciences and crosses disciplines

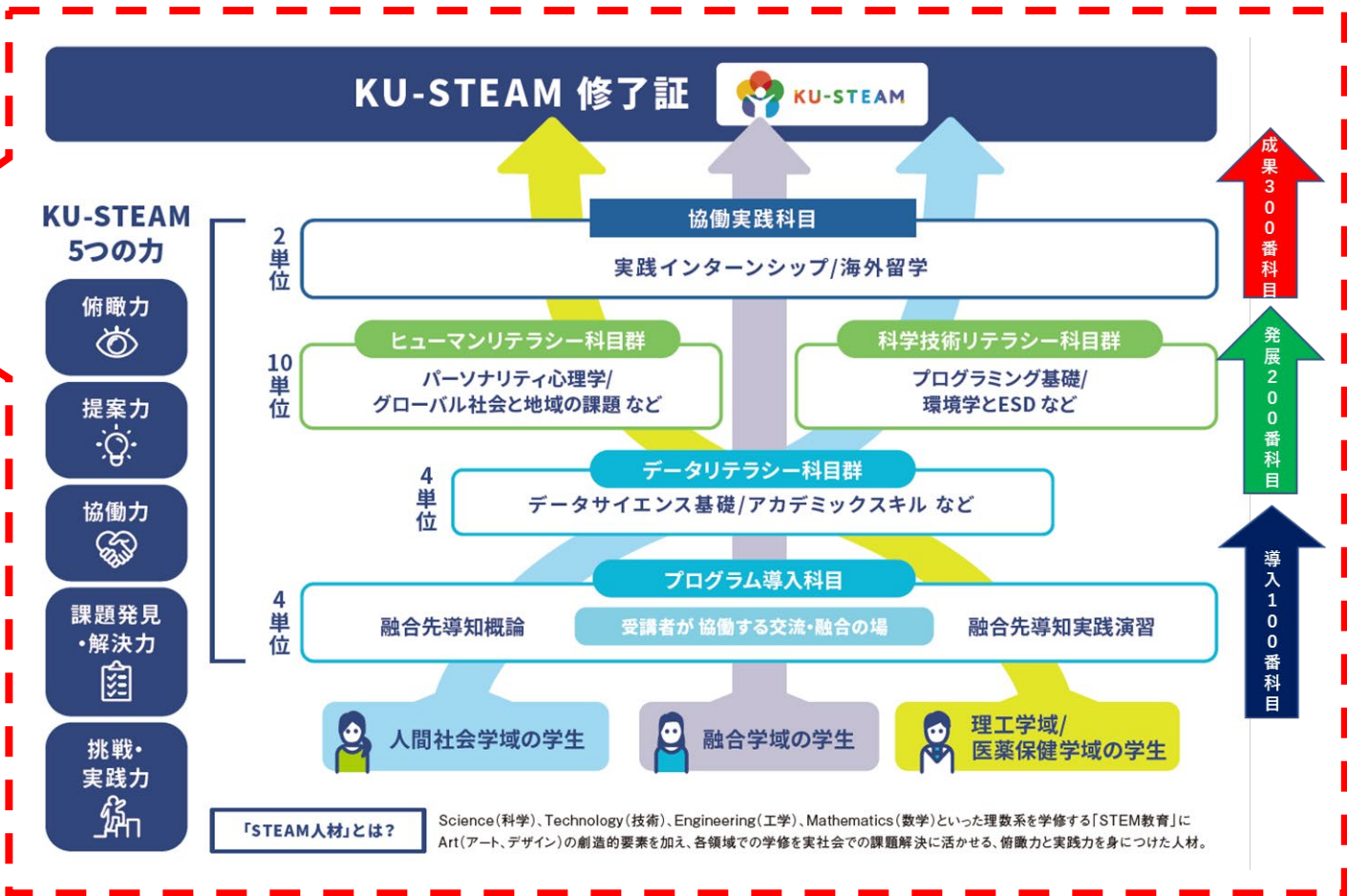
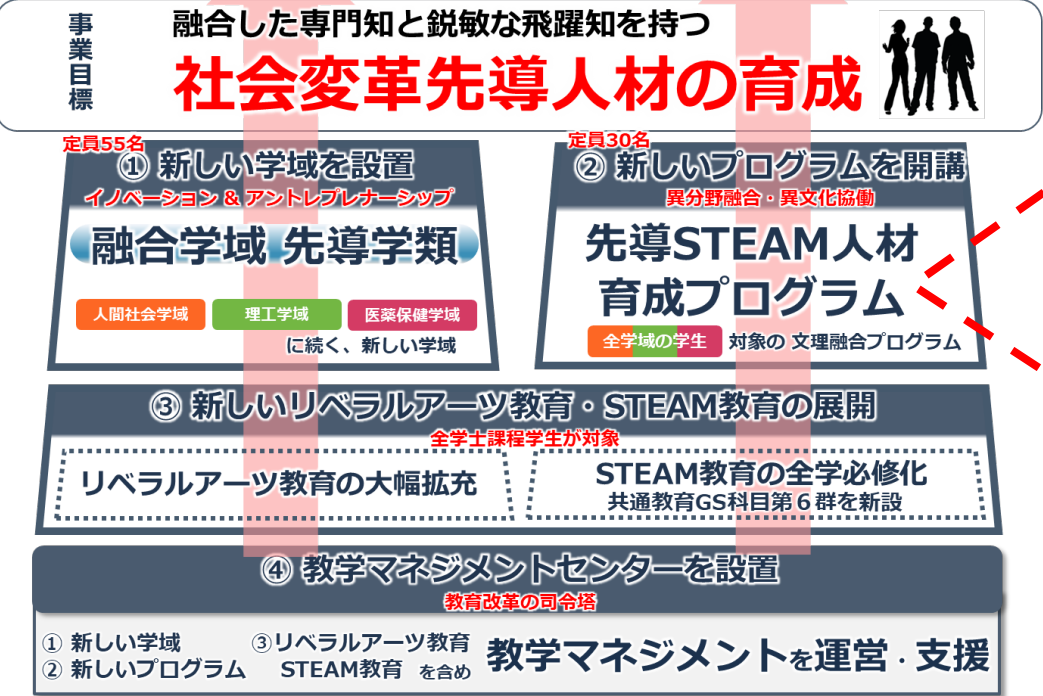
(Including minutes of the first meeting of the Subcommittee for University Promotion of the Central Council for Education and the second meeting of the Subcommittee for University Promotion.)

- (1) It is relatively easy for science students to learn humanities fields, but it is not easy for humanities students to learn science fields.
The challenge that there are many barriers
- (2) Students themselves have difficulty in gaining a sense of conviction and confidence in what they have learned through education that integrates the humanities and sciences and crosses disciplines.
The challenge is that there are
- (3) The issue that it is unclear whether the company will receive a solid evaluation from society and the companies where the students will work.
- (4) Connectivity issues with graduate school education after receiving an education that integrates the humanities and sciences and crosses disciplines in the bachelor's program.

Organizational Efforts in STEAM Education at Kanazawa University (1)

"Building an Ecosystem of High School-University-Graduate School Connections Centered on Inquiry and STEAM Education"

【本事業で取組む4つの実施項目】



Organizational Efforts in STEAM Education at Kanazawa University (1)

"Building an Ecosystem of High School-University-Graduate School Connections Centered on Inquiry and STEAM Education"

Development concept of core subjects (assessment subjects) arranged in three steps: introduction, development, and outcomes.

Introduction No. 100] *R3 has already been developed.

A group (classmates) across the arts and sciences and disciplines

[2] Development No. 200] *R4 Development

Issues and solution approaches that integrate the humanities and sciences and cross-disciplinary fields

[3] Outcome No. 300] *R4 Development

Challenges and practices (experiences) in the fusion of arts and sciences and across disciplines

Depth of Learning



(iii) Result No. 300

Fusion of Humanities and Sciences and Cross-disciplinary Problem Solving Challenge and practice

bachelor's degree thesis graduate school employment

(2) Development No. 200

Issues in the fusion of humanities and sciences and across disciplines and solution approaches, touching on the Mastery, try it out



(1) Introduction No. 100

Creating a place to learn in a group that integrates the humanities and sciences and crosses disciplines



breadth of learning

Organizational Efforts in STEAM Education at Kanazawa University (1)

"Building an Ecosystem of High School-University-Graduate School Connections Centered on Inquiry and STEAM Education"

Upstream" area of the ecosystem

[Creation of an environment for mutual collaboration in inquiry-based learning with high schools in Ishikawa and other prefectures].

Collaboration with Kanazawa University Senior High School and Izumigaoka Senior High School (WWL project, SGH project related schools)

Collaboration with high school student support organizations in the Kaga and Noto areas



Downstream" area of the ecosystem

Development of development and achievement courses with companies, local governments, etc.]

Development of STEAM education utilizing issues faced by companies, local governments, etc.

Development of practical internships centered on design thinking in line with the needs of companies, local governments, etc.



Creating a place with the theme of "Exploration and STEAM (Exploration and STEAM Festa)".

Based on networking through curriculum development in STEAM education, the theme of inquiry and STEAM

Providing placemaking as a university

Organizational Efforts in STEAM Education at Kanazawa University (2)

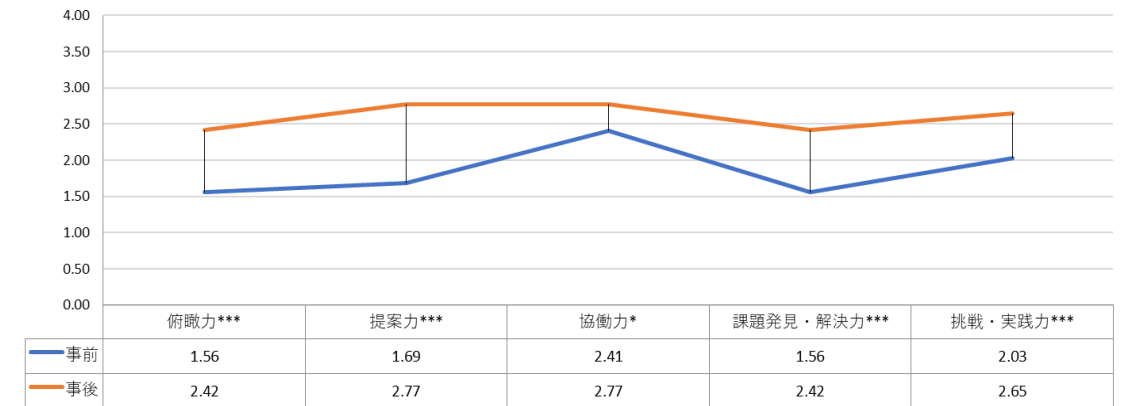
"Assessment that enables verbalization of a sense of growth through fusion of arts and sciences and cross-disciplinary studies".

Core Courses (Introduction to Integrated Leading Knowledge and Practical Exercises for Integrated Leading Knowledge) Level

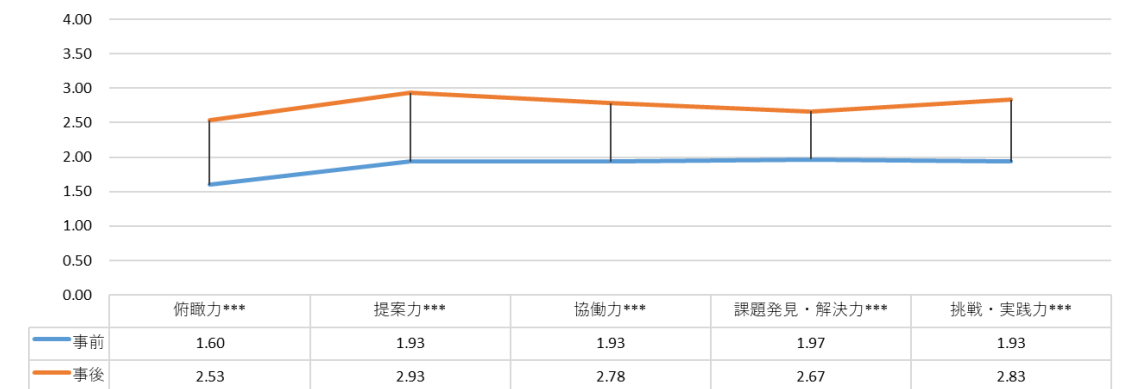
Rubric on the five KU-STEAM competencies

育成する力	説明	Level 1 受動的な態度	Level 2 通常の態度	Level 3 積極的な態度	Level 4 創造的な態度
融合した専門知 (俯瞰力)	個々の能力・適性に合った専門的な知識とともに、幅広い分野や考え方を俯瞰して、自らの判断をまとめ表現できる基盤となる力	○提供された資料の基礎的な事柄を理解。 ○与えられたテーマを調べ、自分の言葉で説明。	○与えられたテーマ以外で自ら関連づけて調べ、自ら資料を入手。 ○相互の比較を交えて専門用語を使って説明。	○多くのことに興味・関心を持ち、複数の専門分野から資料を入手・調査。 ○特徴・課題を分類、比較検討し、専門分野の観点から特色を定性的に理解。 ○自分の考えを複数の専門分野の知識を使って表現	○複数の分野の様々な情報を収集して、与えられたテーマの特色を定量的に理解。 ○数学的・社会科学的手法で課題の抽出・分析・整理し、それに対する自分の考えを自分の言葉で表現。
鋭敏な飛躍知 (提案力)	新しい原理・現象の発見・解明や非連続な技術革新の源泉となる知識を鋭敏に創造できる基盤となる力	○与えられた課題に興味を持ち、修得した知識、基本的な学力を整理 ○従来の技術・伝統を深く知ろうとする	○興味を持ったことを自ら調べ、問題解決の手法を理解し、その特徴を活用。	○学んだ手法を活用して様々なアイデアを自分の言葉で表現。 ○そのアイデアを系統的に整理して、新しい試みを提案。	○従来のものとは異なる新しいアイデアを生み出す。 ○それを活用して新しいものを作り上げる計画を立て、実行に移すためのビジョンを提案。
協働力	目標を共有し、その達成に向けて、お互いの能力・立場を理解し合いながら、コミュニケーションを取り、チームメンバーで役割分担し、協働できる力	○基本的なあいさつ・マナーを身につける。 ○チームの中で自分と与えられた役割を理解し他人の話を聞ける。	○場面に応じた対応ができ、チーム全体の統率を図ろうとする意欲を持つ。 ○その姿勢を自分の言葉で表現。	○必要な場面で自分の考えを伝える姿勢。 チームでの自らの役割を忠実にこなし、チームの結束を促す。	○チームがより効率的に機能するための方法や各構成員が果たすべき役割を理解。 ○相手に対して自分の考えを説明して、具体的な行動に促す。
課題発見・解決力	問題に対して自らの知識や技能を駆使して課題を抽出し、解決のためにその課題に向き合う力	○与えられた情報ツールを活用して課題を抽出。 ○過去の課題解決の事例へのアプローチや解決へのプロセスを理解。	○課題解決のため、必要な情報を理解し、複数の情報を分類・整理。 ○自らの関心ある分野に関し、複数の提案ができる。	○収集した情報を分類・整理し、ミクロ・マクロの視点から、その関連性を抽出。 ○説得力ある根拠とともに、複数の提案ができる。	○収集した情報の重要性・関連性を理解。 問題のメカニズムの可視化や不特定多数に対するわかりやすいプレゼンテーションができる。
挑戦・実行力	身につけた専門知識や技能を様々な場面で活用しようとする積極性。実践に移す行動力	○課題解決のため、自らの置かれている状況、自分が何をすべきかを理解し、取るべき態度を自分の言葉で表現。	○課題解決に自ら当たろうとする姿勢を示す。 ○与えられた役割を理解して、実行に移す。	○課題解決に向けて、自らの役割や取組む分野を限定せず、必要に積極的に関与しようとする姿勢を示す。	○課題解決に向けて、自ら率先して行動することにより、周囲の者も行動へと促す波及効果を示す。

2022年度Q2「融合先導知概論A（サービスマーケティングとアントレプレナーシップ）」
KU-STEAMループリックによるアセスメント結果（N=28）



2022年度Q2「融合先導知実践演習A（ちょこっとマイプロジェクト）」
KU-STEAMループリックによるアセスメント結果（N=30）



Note: *p<.05, ***p<.001

Organizational Efforts in STEAM Education at Kanazawa University (2)

"Assessment that enables verbalization of a sense of growth through fusion of arts and sciences and cross-disciplinary studies".

Creating opportunities to verbalize a sense of growth ("What Learning in Arts and Science Education Means to Me" (video))



2nd year student (1st term) in the Department of Interdisciplinary Studies, Division of Frontier Sciences
*KU-STEAM Leading STEAM Human Resource Development Program (KU-STEAM) students

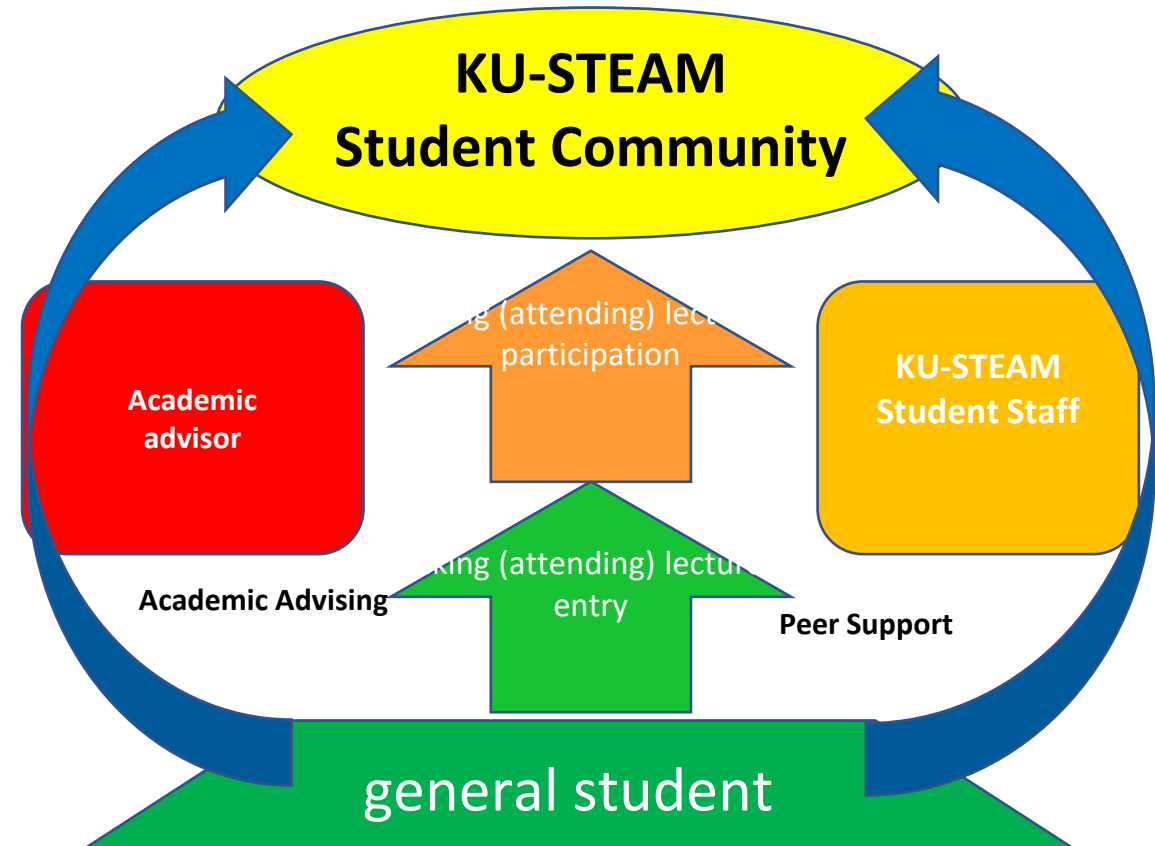


Second-year students of the School of Health Sciences, Faculty of Medicine and Health Sciences
*KU-STEAM Leading STEAM Human Resource Development Program (KU-STEAM) students
*KU-STEAM student staff (peer supporters)

Organizational Efforts in STEAM Education at Kanazawa University (3)

"Academic Advising and Peer Support Integrated Learning Support"

- ◆ Leading STEAM Human Resource Development Program course guide, presentation of course models, etc.
(including public relations activities)
- ◆ Guide to learning opportunities in regular and extra-curricular activities and promotion of growth, etc.
(Autonomous learning support)
- ◆ Creating a chain of connection and community among learners, etc.
(Community Creation Support)



"Academic Advising and Peer Support Integrated Learning Support

Creating a learning community that "learns and grows by itself". (Virtuous Circle through the Student Staff System)

KU-STEAM学生スタッフ募集！

KU-STEAMとは？

金沢大学の学域・学類を超えて、多様な専門知の融合により課題解決力を鍛える分野横断型の先導STEAM人材を育成するプログラムです。





STEAMとは、Science（科学）、Technology（技術）、Engineering（工学・ものづくり）、Art（芸術・リベラルアーツ）、Mathematics（数学）の5つの単語の頭文字を組み合わせた教育概念のことで、技術革新が進み人工知能の影響で世の中が大きく変化する中で生まれました。

■活動内容－予定－

- ・学生同士で学び合うピア・サポート
- ・学生向け説明会等でのプレゼンテーション
- ・共創型ワークショップや実践インターンシップの企画づくり
- ・STEAMラボでのデザイン創作（動画編集、ニュースレター作成）



これから新しく始まる取り組みです。
まずは気楽におしやべりませんか？
担当者より

こんな人におすすめ！

- ・動画編集など新しい挑戦や実績をつくりたい人
- ・他学域・学類の学生と交流し、活動してみたい人
- ・後輩学生たちの学びのサポートに関心がある人

お問い合わせ

KU-STEAM学生スタッフとして参加したい方、関心のある方は、お気軽に下記担当までご連絡ください。

担当：金沢大学 教学マネジメントセンター 山下
メールアドレス：ku-steam@ml.kanazawa-u.ac.jp
（場所：角間キャンパス・インキュベーション施設3階「STEAMラボ」）

Day1
「自分」について知ろう！ → 「ちょこプロ」を考えてみよう！

2022.6.18(土) 9:30-17:30
 ・先輩の人生グラフを聞く
 ・人生グラフの作成
 ・will/can/needの流し出し

6.19(日) 9:30-17:30
 ・ちょこプロ検討ワーク
 ・ちょこプロ宣言

6.20 > 7.22
 アクションしてみよう！ → Day3
 ネクストアクションを考えてみよう！

7.23(土) 9:30-17:30
 ・ちょこプロの振り返り
 ・ネクストアクション検討
 ・プレゼンテーション

▶ プログラムのポイント

- 1 自分のペースで実行してみよう
 ・進捗はグループにシェアしてみよう
- 2 やってみたいことを形にできる
- 3 学域・学年を超えた繋がりができる

▶ サポート者紹介

「やってみたいことが明確でなくても大丈夫」受講生や学生スタッフ、教職員がアイデアを出し一緒にに行きます

金沢大学 教養学部「まちかたへ」 遠 敦哉
 行動して、初めて分かる自分の可能性。みんなと一緒に、自分の可能性を見つけよう！

mummu 代表/金国高校生セイブの「子猫井県地域かん」-いー村上 純一郎
 みなさんと対話しながら、少しずつ楽しい時間を作っています！

金沢大学 理工学域 地球社会基盤学域2年 田村 美波 さあ
 ちょこプロという場とみんなと一緒に楽しみたいのです！

金沢大学 教養学部「まちかたへ」 山田 貴弘 特任助教
 自分と休日のつってみたいことに少し手を付けてみませんか？

金沢大学 医療保健学域 保健学域2年 有賀 一矢 さあ
 やってみたいことも、私と一緒にやってみたい？皆さんと一緒に楽しもうと思えます！

金沢大学 理工学域 地球社会基盤学域2年 伊藤 夢乃 さあ
 やってみたいことをちょこプロでもかき出してはもらえたと嬉しいです！一緒に楽しみたいのです！

【本場聖母堂飲食】 教養マナメントセンター 林 浩・山田 貴弘 E-Mail: k-steam@ipc.kanazawa-u.ac.jp

金沢大学 令和4年度第2回全学FD研修会

参加費無料

ピア・サポート を活用した 学修者本位の 教育の実現

〔知識基約型社会を支える人材育成事業〕幹事校企画

2022.

8.8 月 14:00～16:00

Zoomによるオンライン開催

2019年11月に公表された「2040年に向けた高等教育のグランドデザイン（密着）」と密着に、「学修者本位の教育の実現」が大学教育の根幹において強く要求されています。学修者本位の教育の実現には、学修者自身が学習の主体となり、成長や学びを促すことが不可欠であり、その重要な要素として、学生による学びのための学修支援（ピアサポート）が近年注目されています。特に、本学では、「金沢大学未来ビジョン2.0」において「自ら学び・自ら育む」教育環境の整備を掲げていると、文部科学・労働部によるSTEAM教育の取組を進められている。本学は学生と学修者自身が学習しています。

今回の研修会において、ピア・サポートの基礎概念を学ぶとともに、各職群・学修実践においてどのように設定できることがおのずかにいて、場づくりの事例紹介を交えながら、学生本位の学び合いの意識や価値観に関する理解を深める機会を設けることとする。

対象 教職員、学生ほか
定員 100名

【主催】

「融合した専門知と鋭敏な飛躍性を持つ
社会変革先導人材育成プログラム」
運営委員会

【共催】

FD委員会、教務委員会

▶申込方法

以下申込フォームより参加申込ください。申込受付数、定員200名に達するまで受け付けます。定員に達した場合は、先着順・本学職員の優先参加は原則に行いません。

申込期限：2022年8月3日（水）17:00

<https://forms.gle/PxdaxWe3pPWxMJW57>

知識基約型社会を支える
人材育成事業（AIST）

●お問い合わせ

金沢大学学務部学務課学企係

TEL 076-264-5906 E-mail gwaka@edu.kanazawa-u.ac.jp

Organizational Efforts in STEAM Education at Kanazawa University (3)

"Academic Advising and Peer Support Integrated Learning Support

Out-of-school programs, out-of-school activities (Hidden Curriculum)
Identification and exploitation of resources and potential through



extra-curricular program
Gakusei Social Studies Field Trip" recordings and introductory video
*Interviewed and prepared by KU-STEAM student staff



WATCH THE VIDEO OF TED TALK 2011 BY SALMAN KHAN

Click on the URL below to watch the video (about 20 minutes).

<https://youtu.be/gM95HHI4gLk>



2ND LESSON THEME

"SIGNIFICANCE AND VALUE OF STEAM EDUCATION AND
INTEGRATED KNOWLEDGE IN A KNOWLEDGE INTENSIVE
SOCIETY"

PART 2: SIGNIFICANCE AND VALUE OF INTEGRATED KNOWLEDGE

OUTLINE OF THE SCIENCE, TECHNOLOGY AND INNOVATION BASIC PLAN

Outline of the Science, Technology, and Innovation Basic Plan

Recognition of the Current Situation

Changes in the Situation at Home and Abroad

- Beginning of a reorganization of the world order and increasingly intense leadership competition among countries surrounding science, technology, and innovation (STI)
- Manifestation of global agenda threats such as the climate crisis
- Information monopoly by IT platformers and uneven distribution of great wealth

Accelerates

Expansion of the Novel Coronavirus Infection

- Major Changes in the International Community
 - Rapid social changes in order to prevent the spread of the infection and to maintain economic activities
 - Disruption of the supply chain pressing each country to review the sustainability and resilience of its economy
- Rapidly Changing Life in Japan
 - Transition to a new lifestyle such as work-from-home and online education

Review of STI policies

- Digitalization for digitalization's sake and relative decline in research capabilities
 - Digitalization focuses on improving efficiency of existing operations and the original power of ICT is not being fully utilized.
 - Decline of international standing for research papers and severe research environment continues
- Revision of the Basic Act on Science and Technology
 - STI policies should contribute to comprehensive understanding and problem solving of human beings and society through the "convergence of knowledge" that fuses the natural sciences with humanities and social science

Balancing response to global issues with the reform of social structures in Japan is essential

Society That Japan Aims for (Society 5.0)

Sustainable and Resilient Society That Ensures the Safety and Security of the People

[Securing Sustainability]

- Realization of a **sustainable global environment** with a focus on achieving the SDGs
- Realization of a society in which **future generations can live in abundance while satisfying the needs of the present generation**

Incorporate **traditional Japanese values** of trust and sharing into this vision for society and transmit it to the world as **Society 5.0**.

[Securing Resilience]

Realization of **comprehensive security** against threats such as disasters, infections, cyber terrorism, increasingly severe security environment, and disruption of the supply chain

What is Necessary to Realize Society 5.0

Transformation into a sustainable and resilient society through the fusion of cyberspace and physical space

A Society in Which Each Individual Can Realize Diverse Happiness (Well-Being)

[Realization of Economic Affluence and Qualitative Affluence]

- Realization of an educational, labor, and employment environment that enables everyone to **develop their own abilities and diverse work styles** that utilize such abilities
- Realization of an environment that allows people to **participate in society in health throughout their life** in an age of a 100-year lifespan
- Realization of a society that allows people to continue to have their dreams and always **participate in society with a positive view of their presence**

Contribute to the international community and attract global **human resources** and **investment**

Creation of "knowledge" as a source of value creation by designing a new society

Development of human resources to support a new society

Push through **social transformation** and advance **investment** looking ahead into the **future (knowledge and human resources)**

STI Policy for the Realization of Society 5.0

- Draw up policies based on **backcasting** from the future vision and **forecasting** from the current situation while utilizing **convergence of knowledge** and **evidence**, and flexibly improve them through evaluation.
- Aim for a total government R&D investment of approximately **30 trillion yen** and a total public and private R&D investment of approximately **120 trillion yen**.

Transformation into a sustainable and resilient society that ensures the safety and security of the people

- (1) Creation of new value through the **fusion of cyber space and physical space**
 - Digitalizing the government, launching a Digital Agency, and completing a data strategy (developing a base registry, etc.)
 - Maintaining and developing next generation infrastructure and technologies for Beyond 5G, supercomputers, space systems, quantum technologies, semiconductors, etc.
 - (2) Advancement of social changes and discontinuous innovation aimed at **overcoming issues on a global scale**
 - Promoting R&D (utilizing funds, etc.) and reducing costs of innovative environmental innovation technologies and transitioning to a circular economy
 - (3) Building of a **resilient, safe and secure society**
 - Identifying and R&D of important technologies for responding to threats and advancing social implementation and technology outflow countermeasures.
 - (4) Formation of an **innovation ecosystem** that is the foundation for creating new value-sharing industries
 - Advancing an SBIR system and entrepreneurial education, forming start-up hub cities, and strengthening a co-creation system through industry, academia, and government collaboration
 - (5) Urban and regional development (development of **smart cities**) as the foundation for succeeding to the next generation
 - Creating smart cities and super cities, their nationwide spread through a public-private collaboration platform, and international deployment at expos.
 - (6) R&D for solving various social issues, advancement of **social implementation**, and utilization of **convergence of knowledge**
 - Social implementation through the utilization of convergence of knowledge, review and formulation of evidence-based national strategies, and advancement of R&D
 - Advancing SIP and moonshot R&D, market gain through the utilization of intellectual property and standards, and advancing new science and technology diplomacy
- AI technologies; biotechnologies; quantum technologies; materials; space; ocean; environmental energy; health and medical care; food; agriculture, forestry, and fisheries; etc.

Demand from society

Injection of knowledge and human resources

Development of frontiers of knowledge and strengthening research capabilities as sources of value creation

- (1) Rebuilding the environment to produce **diverse and outstanding research**
 - Improving the treatment of doctoral students and expanding their career paths and securing posts for young researchers
 - Promoting active participation of female researchers, strengthening basic research and academic research, and advancing joint international research and international brain circulation
 - Strengthening the humanities and social sciences and creating convergence of knowledge (strengthening funding and DX of research in the humanities and social sciences)
- (2) Construction of new research systems (promotion of **open science** and **data-driven research**, etc.)
 - Managing and utilizing research data and acceleration of research utilizing smart labs, AI, etc.
 - Maintaining and sharing research institutions, facilities, and equipment and fostering of new research communities and environment cultivated by research DX
- (3) Promotion of **university reform** and expanding functions for **strategic management**
 - Developing diverse and unique university groups (transition to a true management entities and further growth as research universities that are on par with top international universities)
 - Creating a 10 trillion yen university fund

Education and human resource development to realize diverse happiness for each individual and ability to face challenges

Transition to an education and human resources development system that enhances people's **ability to explore and attitude to continue learning**

- Advancing STEAM education from the elementary and secondary education stage and the GIGA School Concept, and reducing teachers' burden
- Providing diverse curricula and programs at universities, etc. and fostering an environment and culture that promotes recurrent education

THE SCIENCE, TECHNOLOGY AND INNOVATION BASIC PLAN

➡ Please read Material 2
on WebClass

[Tentative Translation]

Science, Technology, and Innovation
Basic Plan

March 26, 2021
Government of Japan

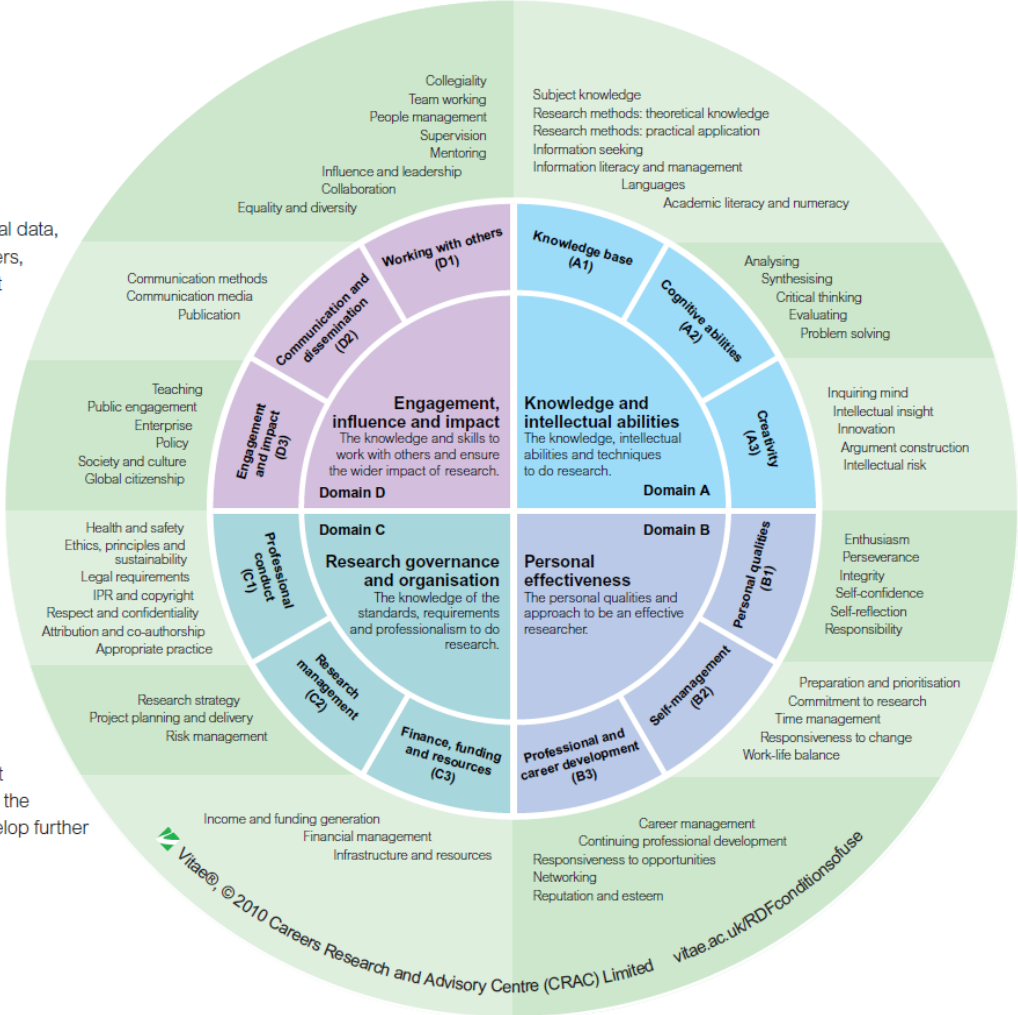
Transferable Skills Required of Graduate Students Transferable Skills
Research Development Framework as a reference (2010, Vitae, UK)

(Source: <https://www.vitae.ac.uk/vitae-publications/rdf-related/researcher-development-framework-rdf-vitae.pdf/view>)

Structure

The RDF has been created from empirical data, collected through interviewing researchers, to identify the characteristics of excellent researchers expressed in the RDF as 'descriptors'. The descriptors are structured in four domains and twelve sub-domains, encompassing the knowledge, intellectual abilities, techniques and professional standards to do research, as well as the personal qualities, knowledge and skills to work with others and ensure the wider impact of research. Each of the sixty-three descriptors contains between three to five phases, representing distinct stages of development or levels of performance within that descriptor.

The RDF has been incorporated into a downloadable Professional Development Planner to enable researchers to identify the areas in the framework they want to develop further and to create an action plan.



2ND LESSON SUBMISSION



Second class assignment (to be submitted via WebClass)

After reading some handout materials and viewing the on-demand videos, "Please state your own thoughts while explaining (1) the significance and value of STEAM education and (2) the significance and value of comprehensive knowledge."

The entire application must be between 300 and 400 words and submitted via WebClass by a date to be informed.

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REFERENCES

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